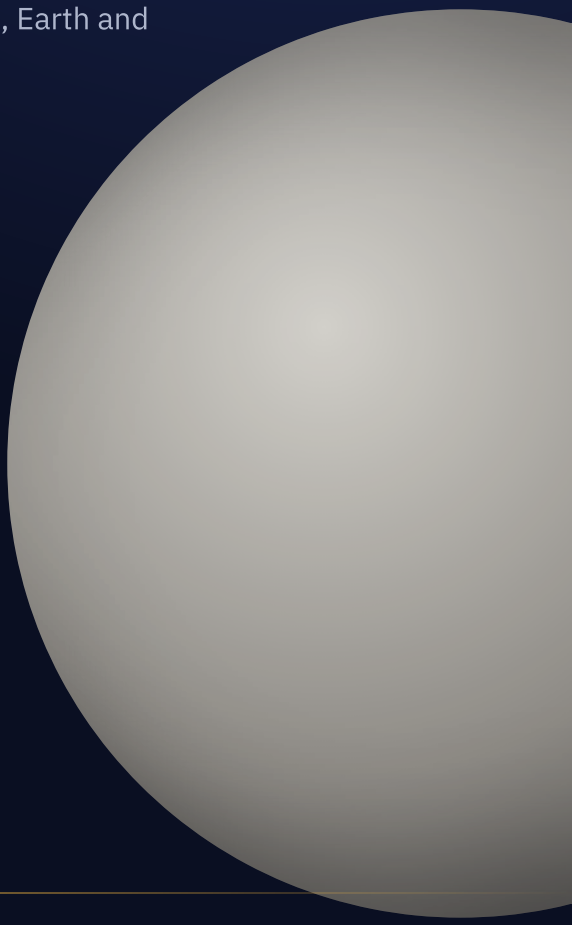
NASA SPACE APPS CHALLENGE 2026 · LUNAR LANDING SITE NAVIGATOR

LunaHorizon

Software Requirements Specification

Find the right place and the right time to land on the Moon: Sun, Earth and terrain visibility for any lunar landing site and date.



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VERSION
1.0 · Draft for mentor review

DATE
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REPOSITORY
github.com/Mozahid-DIU/LunaHorizon (MIT)

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1 Introduction

1.1 Purpose

This document specifies what LunaHorizon must do and how well it must do it. It is the reference for building, testing and judging the product, and is written for the team, our Space Apps mentor and reviewers.

1.2 Problem

At the lunar south pole the Sun stays within a few degrees of the horizon and Earth rises and sets from view. Crater rims and mountains can block both. A lander that picks the wrong site or date can lose solar power, direct-to-Earth communication, or both. Accurate tools exist (for example JPL Horizons), but their tabular output needs an expert to interpret quickly.

1.3 Scope

LunaHorizon is a public web application. For a curated or custom landing site and a date between 2026 and 2030, it shows where the Sun and Earth are against the real terrain horizon, when power and communication windows open and close, how long they last, which landing dates are best, and how results change when a launch slips or the landing point moves.

> **In scope:** Sun and Earth geometry, terrain horizons, power and communication windows, DSN coverage, landing-ellipse risk, a power and battery estimate, 3D terrain shadows, comparison, plain-language summaries, validation against JPL Horizons.
>
> **Out of scope:** thermal modelling beyond simple hints, slope and hazard analysis for touchdown, orbital and trajectory design, relay satellites, real-time telemetry.

1.4 References

- NASA Space Apps Challenge 2026: *Lunar Landing Site Navigator* challenge statement and participant guide
- NASA JPL Horizons System API · NAIF SPICE kernels (DE440, Moon PA DE421, Moon frame kernels)
- LRO LOLA polar DEMs (PDS Geosciences Node) · SLDEM2015 (LOLA + JAXA SELENE/Kaguya)
- NASA Deep Space Network station locations · NASA CLPS and mission pages for site coordinates

2 Overall description

2.1 Product perspective

LunaHorizon is a standalone, open-source web app. Heavy computations run once, offline, in a Python pipeline; results are shipped as static JSON. The browser performs all per-site calculations, so the app needs no database and no general backend. Two small serverless functions provide AI explanations and question parsing, each with a rule-based fallback.

2.2 User classes

USER	GOAL	WHAT THEY NEED MOST
Mission planner / engineer	First-pass check of sites and dates	Accurate numbers, windows, best dates, slip, ellipse risk, power, export
Scientist	Judge how long a site supports operations	Golden-window duration, compare, mission brief
Educator / student	Understand why the south pole is hard	Sky View, 3D view, Mission Replay, Classroom mode, plain language
Public	Follow Artemis and CLPS missions	Simple visuals, share links, works on a phone

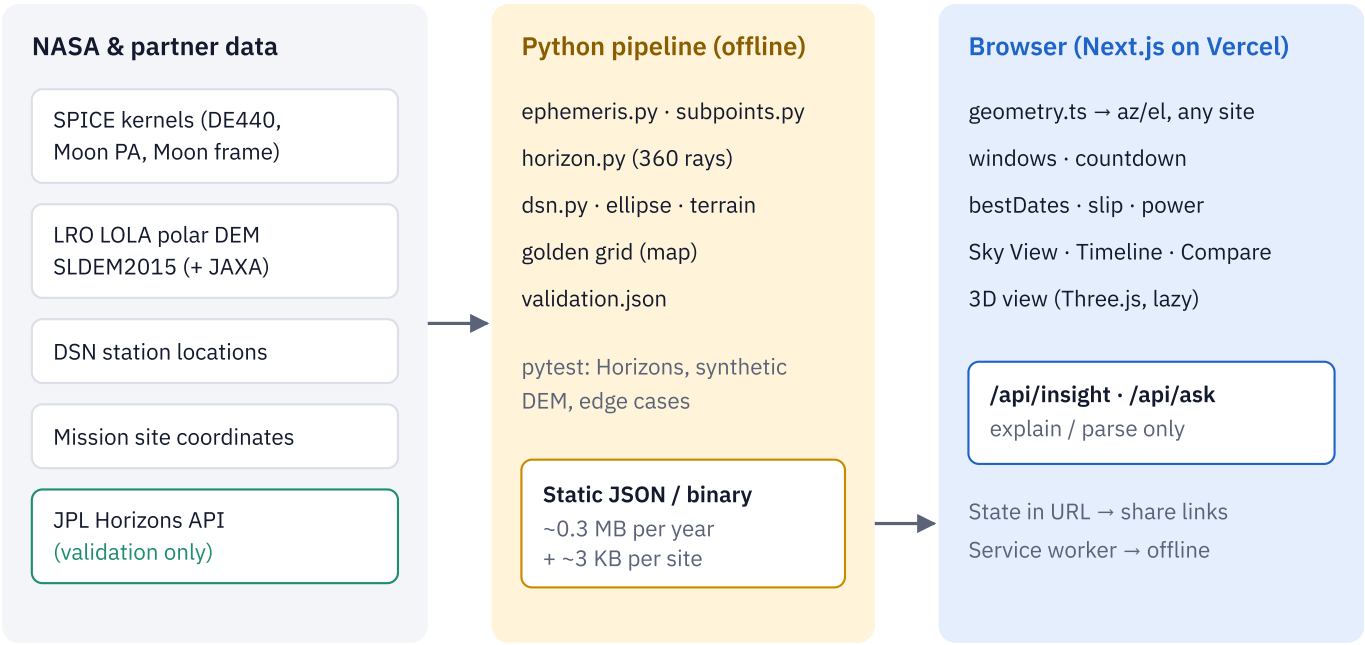
2.3 Operating environment

- Current Chrome, Edge, Firefox and Safari on desktop and mobile; minimum width 360 px
- Hosted on Vercel; installable as a PWA for offline use
- Pipeline: Python 3.11 on Windows, macOS or Linux

2.4 Constraints

- Open source (MIT), public repository, as required by Space Apps
- No NASA logos, flags or mission insignia; AI use disclosed in `docs/ai-use.md`
- AI must never compute positions, windows or power. It may only explain or parse
- Built by a small team in 15 days; development before the event approved by the team's mentor

3 System architecture



3.1 Key design decisions

DECISION	REASON
Precompute hourly sub-solar and sub-Earth points (2026–2030) once	One small file serves every curated and custom site; no server needed
Per-site az/el, windows and power computed in the browser	Instant results (one year = 8,760 hours); custom coordinates work everywhere
Same logic mirrored in Python	Browser output is tested against the validated Python engine
All state in URL parameters	Any view can be shared or bookmarked; no accounts
AI behind a rule-based fallback	The demo never breaks without internet or an API key

4 Calculation model

4.1 Sun and Earth position

Topocentric azimuth and elevation of the Sun and Earth are computed with Skyfield and NAIF SPICE kernels (DE440 ephemeris, Moon principal-axis frame MOON_ME_DE421, which includes physical libration). Light-time and aberration are applied. The Moon has no atmosphere, so no refraction is applied.

4.2 Terrain horizon

For each site, 360 rays (1° steps) are cast over the DEM out to 150 km. The horizon elevation in each direction is the largest angle to any terrain point, corrected for lunar curvature:

```
h(az) = max over d of  atan( (z(d) − z0 − d² / 2R) / d ),   R = 1737.4 km
z0 = site elevation + lander height (default 2 m)
```

4.3 Windows

WINDOW	RULE (HOURLY, INTERPOLATED TO MINUTES FOR COUNTDOWNS)
Lit	Sun elevation − h(Sun azimuth) > +0.27°; partial between −0.27° and +0.27° (fraction of solar disc visible)
Earth visible	Earth elevation > h(Earth azimuth)
Comms (DSN-aware)	Earth visible **and** the Moon is above 10° at Goldstone, Madrid or Canberra
Golden	Lit **and** Comms at the same time
Best dates	For each landing hour, total Golden hours over the mission length (default 14 days); top 5 shown
Slip	Same score at +1, +7 and +30 days and at nearby sites, shown as a change from the baseline

4.4 Power and battery

```
P(t) = S · (1 AU / r(t))² · A · η · max(0, cos θ(t)) · f(t)
SoC(t+1) = clamp( SoC(t) + (P(t) − L) · Δt ,  0 ,  C )

S = 1361 W/m²   r = Sun distance   A = panel area   η = efficiency
θ = angle between Sun and panel normal   f = visible fraction of solar disc
L = load (W)    C = battery capacity (Wh)    Δt = 1 h
```

The lander "survives the night" if SoC never reaches 0 during the mission. This is an engineering estimate for comparison, not a flight power model.

4.5 Landing ellipse

Horizons are precomputed at 25 points inside a 1 km radius around each demo site. For a user-selected ellipse (100 m – 1 km), the app reports best, median and worst lit percentage and Golden hours over the points that fall inside it.

5 Functional requirements

Priority: `MUST` required for the demo · `SHOULD` planned, cut only if time runs out · `COULD` if time allows.

5.1 Core (Tier 1)

ID	REQUIREMENT	PRIORITY	ACCEPTANCE
FR-01	The user can pick a site from ~20 curated sites (Apollo, CLPS, Chandrayaan-3, Chang'e, SLIM, Artemis III regions) or enter custom latitude and longitude.	MUST	Each curated site shows its source; invalid lat/lon rejected with a clear message
FR-02	The user can pick a date and time (UTC) between 2026-01-01 and 2030-12-31.	MUST	Out-of-range dates are blocked
FR-03	**Sky View:** a 360° panorama showing the terrain horizon, Sun position and path, and Earth position, with play/pause animation over a lunar day.	MUST	Sun and Earth markers match engine az/el; animation runs at ≥ 30 fps
FR-04	**Countdown:** time to next sunset, Earthset, sunrise and Earthrise, plus current solar elevation.	MUST	Matches window boundaries to the minute
FR-05	**Timeline:** 30-day bars for lit, comms and Golden windows with durations.	MUST	Bar edges equal computed window edges
FR-06	**Compare:** up to 3 sites or dates side by side.	MUST	Same metrics shown for each column
FR-07	**Best dates:** top 5 landing dates in the next 6–12 months for a chosen mission length.	MUST	Ranking equals Python reference
FR-08	**Slip Planner:** change in Golden hours for +1, +7, +30-day slips and nearby sites.	MUST	Deltas shown with sign and %
FR-09	**Validation Panel:** our Sun and Earth az/el next to JPL Horizons for reference sites and times, with the difference.	MUST	All differences < 0.1°
FR-10	**Insight:** 2–3 sentence plain-language summary of the current results.	MUST	Only quotes computed numbers; rule-based text when AI is unavailable

5.2 Mission-grade features

ID	REQUIREMENT	PRIORITY	ACCEPTANCE
FR-11	**DSN-aware comms:** comms requires Earth visible and at least one DSN complex seeing the Moon; the station is named.	MUST	Timeline shows "Earth up" and "Earth up + DSN" separately
FR-12	**Landing Ellipse Risk:** best, median and worst lit % and Golden hours inside a 100 m – 1 km ellipse.	MUST	Results update when the ellipse size changes
FR-13	**Power & Battery Simulator:** inputs panel area, efficiency, orientation, battery, load; outputs hourly state of charge and a survive-the-night verdict.	MUST	Known reference case gives the expected verdict
FR-14	**Lander's-eye 3D view:** LOLA terrain for 3 sites with Sun lighting and moving shadows.	SHOULD	Shadow reaches the site within 1 h of computed sunset
FR-15	**Ask LunaHorizon:** a plain-English question is converted to engine parameters and answered by the engine.	COULD	Falls back to a filter form

ID	REQUIREMENT	PRIORITY	ACCEPTANCE
FR-16	**Mission Brief:** one-click printable one-page summary for the current site and date.	SHOULD	Prints to a single A4 page

5.3 Exploration and learning (Tier 2)

ID	REQUIREMENT	PRIORITY	ACCEPTANCE
FR-17	**Golden Window Map:** south pole heatmap of Golden hours; clicking a cell opens its Sky View.	SHOULD	Click sets site in URL
FR-18	**Small change, big impact:** % change when moving the site 1–2 km or the date by one day.	SHOULD	Shows both absolute and % change
FR-19	**Mission Replay:** the sky at landing for Apollo 11, Chandrayaan-3, IM-1 and IM-2.	COULD	Uses official landing time and site
FR-20	**Classroom mode:** guided lessons using the live tools.	COULD	At least one complete lesson

5.4 Utilities (Tier 3 and extras)

ID	REQUIREMENT	PRIORITY	ACCEPTANCE
FR-21	**Share link:** every view is reproducible from its URL.	MUST	Opening a copied URL restores the same view
FR-22	**CSV export** of hourly az/el and window flags.	COULD	Opens in Excel with headers
FR-23	**Thermal hints:** hours since sunrise / in shadow as a simple indicator.	COULD	Labelled as indicative only
FR-24	**Offline mode** (PWA) for previously opened sites.	COULD	Works in airplane mode after first visit
FR-25	**Open data:** all generated JSON is downloadable and documented.	COULD	`docs/data.md` describes every file

6 External interfaces

6.1 User interface

- One main screen: site + date controls, Sky View, countdown bar, timeline; secondary panels for Compare, Best dates/Slip, Power, Ellipse, Validation, 3D
- Colour meaning is fixed across the app: **gold = Sun / power**, **blue = Earth / comms**, mint = validated / Golden
- Every colour-coded state also has a text label or pattern (colour-blind safe)

6.2 Software interfaces

INTERFACE	DIRECTION	CONTRACT
`GET /data/subpoints-<year>.json`	App ← static	Hourly sub-solar and sub-Earth lat/lon and distances
`GET /data/horizon/<site>.json`	App ← static	360 horizon elevations in degrees, DEM source and resolution
`GET /data/dsn-<year>.json`	App ← static	Hourly bitmask of DSN complexes seeing the Moon

INTERFACE	DIRECTION	CONTRACT
`POST /api/insight`	App → serverless	Input: computed metrics only. Output: ≤ 3 sentences. Fallback on error or timeout (5 s)
`POST /api/ask`	App → serverless	Input: question text (≤ 300 chars). Output: validated parameter JSON, never numbers
JPL Horizons API	Pipeline → NASA	Used only to build validation fixtures; not called by the live app

7 Data requirements

DATASET	PROVIDER	USE
DE440 planetary ephemeris (`de440s.bsp`)	NASA JPL / NAIF	Sun, Earth, Moon positions
Moon PA DE421 + frame kernels	NASA JPL / NAIF	Lunar orientation and libration
LOLA polar DEM (~20 m/px, 87.5°S–90°S)	NASA LRO / PDS	Terrain horizons, ellipse, 3D
SLDEM2015 (~60 m/px, ±60°)	NASA LOLA + JAXA Kaguya	Horizons for non-polar sites
DSN complex coordinates	NASA DSN	Comms coverage
Landing site coordinates	NASA, ISRO, CNSA, JAXA, CLPS providers	Curated sites, Mission Replay
JPL Horizons observer tables	NASA JPL	Validation
Official challenge resources	Space Apps (13 Nov)	Matched and credited on release

Every curated site stores: id, name, mission, agency, latitude, longitude, landing date (if any), source URL. Sites without a terrain tile use a spherical horizon (0°) and are labelled as such.

8 Non-functional requirements

ID	QUALITY	REQUIREMENT
NFR-01	Accuracy	Sun and Earth az/el within **0.1°** of JPL Horizons (achieved: < 0.0001° on 24 checks)
NFR-02	Accuracy	Browser engine within **0.02°** of the Python engine
NFR-03	Accuracy	Terrain horizon: flat DEM gives 0° ± 0.01°; synthetic hill gives the analytic angle ± 0.05°
NFR-04	Performance	LCP < 2.5 s on 4G; result update after changing site or date < 300 ms; best dates for a year < 1 s
NFR-05	Performance	Lighthouse performance and accessibility ≥ 90; core JS < 300 KB gzipped; 3D loaded only on demand
NFR-06	Reliability	App fully usable without the AI service; no single external call can block a view
NFR-07	Usability	A first-time user gets a Sky View and countdown for a site in ≤ 3 clicks; works on 360 px wide phones
NFR-08	Accessibility	WCAG 2.1 AA contrast; full keyboard use; screen-reader labels and text summaries for charts
NFR-09	Security	API key only in server environment; input validated on both client and server; rate limit on `/api/*`; CSP and security headers; no user accounts or personal data

ID	QUALITY	REQUIREMENT
NFR-10	Honesty	Every estimate (power, thermal, ellipse) states its assumptions; data sources and AI use shown in the app
NFR-11	Maintainability	Tests for every calculation (pytest, Vitest); ≥ 80% coverage on engine code; CI on every push
NFR-12	Portability	Pipeline reproducible from `requirements.txt` and `get_kernels.py`; data sources swappable without engine changes

9 Verification and traceability

9.1 Test strategy

LEVEL	TOOL	WHAT IS CHECKED
Reference	pytest + JPL Horizons fixtures	Sun/Earth az/el at equatorial, mid-latitude and polar sites
Unit	pytest, Vitest	Horizon rays, window edges, azimuth wrap, longitude ±180°, year boundaries, battery limits
Parity	Vitest vs Python output	Browser and pipeline give the same numbers
End-to-end	Playwright on the live URL	Pick site → pick date → Sky View, countdown, compare, share link
Quality	Lighthouse, axe	Performance, accessibility
Plausibility	LRO illumination maps, LROC QuickMap	Lit % and horizon shape at Shackleton rim

9.2 Traceability to the challenge

CHALLENGE OBJECTIVE	REQUIREMENTS
Show Sun and Earth position relative to the local horizon	FR-03, FR-09, FR-14
Power generation potential and solar elevation	FR-04, FR-05, FR-13
Direct-to-Earth communication windows	FR-05, FR-11
Countdown to sunset / Earthset as operational information	FR-04
Effect of small changes in location or timing	FR-08, FR-12, FR-18
Compare scenarios and coordinate science and engineering teams	FR-06, FR-16, FR-21
Curated and custom sites	FR-01, FR-17
Accessible to non-experts and education	FR-10, FR-19, FR-20, NFR-07, NFR-08

10 Release plan

GATE	DATE	CONTENT
G0	28 Oct ✅	Ephemeris engine validated against JPL Horizons
G1 · Engine	3 Nov	Sites, sub-points, DSN, terrain horizons, windows, power — all tested
G2 · Core live	8 Nov	FR-01 – FR-10, FR-21 on Vercel; Playwright and Lighthouse pass

GATE	DATE	CONTENT
G3 · Mission-grade	10 Nov	FR-11 – FR-14
G4 · Freeze	12 Nov	Remaining SHOULD/COULD items, or hidden
Event	13–14 Nov	Match official resources, final video, project page submission

11 Risks and assumptions

RISK / ASSUMPTION	MITIGATION
DEM files are large and slow to process	Use only needed tiles; start early; coarse map grid first
DEM resolution limits horizon accuracy near small features	State resolution in the app; use 20 m polar DEM where available
Official resources (13 Nov) add new datasets	Engine is data-agnostic; only sources change
AI service or key unavailable	Rule-based fallback for all AI features
Hosting or internet failure during judging	Local build and offline PWA as backup
Power and thermal models are simplified	Labelled as estimates with assumptions (NFR-10)

A Glossary

Azimuth / elevation	Compass direction and angle above the horizon of an object in the sky
Terrain horizon	The skyline formed by surrounding hills and crater rims, per direction
Sub-solar / sub-Earth point	The point on the Moon directly below the Sun / Earth at a given time
Golden Window	Time when a site has both sunlight and communication with Earth
DSN	NASA Deep Space Network: antenna complexes at Goldstone (USA), Madrid (Spain), Canberra (Australia)
Landing ellipse	The area in which a lander is expected to touch down, given navigation errors
Libration	The Moon's slight wobble as seen from Earth, which moves Earth in the lunar sky
SPICE	NASA NAIF toolkit and data files used by missions for positions and orientations
LOLA	Lunar Orbiter Laser Altimeter on NASA's Lunar Reconnaissance Orbiter; source of elevation maps
CLPS	NASA Commercial Lunar Payload Services program
SoC	Battery state of charge

LunaHorizon is a student hackathon project and is not affiliated with or endorsed by NASA. This document was drafted with AI assistance (Claude) and reviewed by the team.